Complaint Investigation RAG

Built a RAG system that retrieves similar complaint cases and turns them into evidence-grounded investigation support.

Problem Statement

Traditional retrieval systems can return similar complaint cases, but users still need to manually read multiple results and reason through the issue.

This project explores whether RAG can go beyond retrieval and generate investigation-style analytical support grounded in historical complaint evidence.

Dataset

Consumer Complaint Database from Kaggle, which contains historical consumer finance complaints, complaint narratives, issue categories, and company response fields. This makes it suitable for exploring similar-case retrieval and evidence-grounded complaint analysis.

Workflow

1. **Data Cleaning & Structured Complaint Representation**
   Cleaned raw data and organized it into structured case records with complaint text, issue labels, and response fields.

2. **Case-level Vector DB**
   Retrieves full historical complaint cases to preserve overall dispute context.

3. **Chunk-level Vector DB**
   Retrieves more precise evidence chunks to surface the strongest supporting details.

4. **Hybrid Retrieval (Case + Chunk)**
   Combines full-case similarity and evidence-level relevance.

5. **LLM-based Investigation Output**
   Transforms retrieved evidence into investigation-style review support.

6. **Streamlit Demo**
   Provides an interactive interface for exploring the pipeline end-to-end.

Technical Design Choices

1 Data Preparation

- Selected key complaint fields and removed incomplete records
- Standardized complaint text
- Created both case-level and chunk-level retrieval units

2 Retrieval

- **BAAI/bge-small-en-v1.5** : A relatively small embedding model with strong semantic retrieval performance
- **Chroma** vector database

3 Generation

- OpenAI API
- Prompt engineering for investigation-style output : used to transform retrieved evidence into outputs such as similarity pivot

4 Demo

- **Streamlit**: a fast and practical way to turn the pipeline into an interactive demo without building a full front-end stack

From Retrieval to Analytical Support

Traditional Retrieval Output

- Top similar cases
- Evidence text only

My Investigation-Style Output

- Likely Issue Category
- Core Conflict Point
- Similarity Pivot
- Risk Alert
- Suggested Review Checks

Retrieval Evaluation

Retrieval Type	Precision@3	Precision@5	Hit Rate@3	Hit Rate@5	NDCG@3	NDCG@5
Case	0.978	0.587	1.0	1.0	0.980	0.980
Chunk	0.956	0.960	1.0	1.0	0.879	0.941

Evaluation Setup

- Initial relevance labels were created manually for a seed set of query–result pairs
- LLM-assisted labeling was used to expand the evaluation set
- Final labels were then used to compute retrieval metrics

Demo

Complaint Investigation RAG

Historical case retrieval + investigation-style analysis

Use example 1 | Use example 2

Enter a new complaint

A merchant charged my credit card twice by mistake

Analyze Complaint

Investigation Summary

Likely Issue Category

Duplicate merchant charge — missing merchant authorization proof

Core Conflict Point

The case turns on whether the merchant can produce documentary proof (signed receipts, distinct authorization IDs/order/guest check numbers, or POS logs) showing two valid, separately authorized charges; if the merchant cannot, the chargeback/correction path is supported, but if the merchant provides apparent independent authorizations the issuer may justify denying reversal or re-billing temporary credits.

Risk Alert

The single biggest blocker is absence of merchant authorization records (signed receipts, unique auth IDs, or separate guest-check numbers).

Similarity Pivot

Closest match: complaint_id 5117215 — both describe a split-bill scenario where the customer was charged twice for the same amount on the same date and the merchant/issuer response relied on the merchant's representation of 'two separate bills' without providing signed receipts.

Likely Review Paths

Path A: Issuer finds merchant cannot produce distinct authorization evidence (signed receipts or separate auth codes) and issues monetary relief/chargeback to the consumer. Path B: Merchant provides documentation purporting to show two valid charges or issuer defers to merchant records, resulting in dispute denial or reversal of a provisional credit.

Actionable Checklist

To do	Confirm both charges on the account: compare amounts, merchant descriptor, transaction timestamps, transaction IDs and authorization codes from the statement/processor feed.
To do	Check issuer dispute history and notes for this card/account: any prior disputes, provisional credits issued, reversals, and timestamps of those actions.
To do	Request merchant POS logs/guest checks and signed receipts for both transactions, including transaction/authorization codes and terminal IDs.
To do	Obtain merchant's formal explanation tying each charge to a separate order (order/guest check numbers, server notes) and any admissions of error or refund authorizations.
To do	If merchant cannot produce distinct authorizations, prepare chargeback package: consumer affidavit, transaction metadata, merchant non-response documentation, and issuer dispute submission notes.

This is a historical investigation reference, not a guaranteed resolution.

Top Similar Cases

Case 1

Score: 0.3735

Complaint ID: 5117215

Product: Credit card or prepaid card

Issue: Problem with a purchase shown on your statement

Company: ZIONS BANCORPORATION

Company Response: Closed with monetary relief

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Product: Credit card or prepaid card
Sub-product: General-purpose credit card or charge card
Issue: Problem with a purchase shown on your statement
Sub-issue: Credit card company isn't resolving a dispute about a purchase on your statement
Complaint: I have contacted the credit card company and the merchant over 10 times going back to XXXX of 2021 trying to resolve the issue. I expl
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Thank You